

Notes:

- The GBN protocol refers to Go-Back-N protocol
- For P.15, assume the bandwidth and the one-way propagation delay of the cross-country link is 100Mbps and 25ms, respectively.
- For P.37, only consider the comparison between GBN and TCP protocols only (do not consider SR protocol)
- The following figure is referred to by the questions:

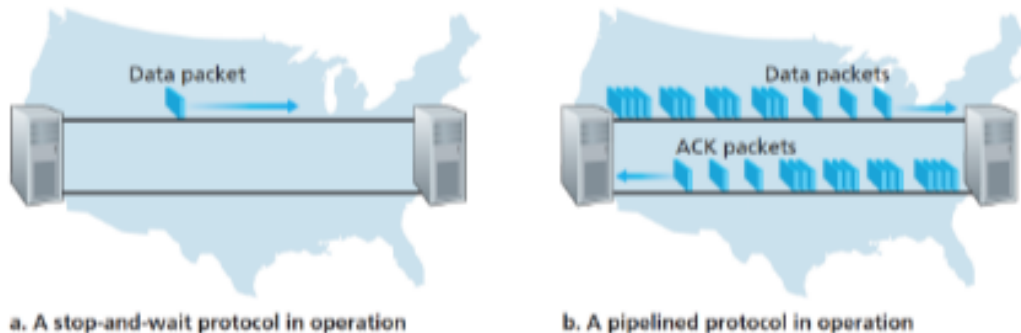


Figure 3.17 ♦ Stop-and-wait versus pipelined protocol

- P15. Consider the cross-country example shown in Figure 3.17. How big would the window size have to be for the channel utilization to be greater than 98 percent? Suppose that the size of a packet is 1,500 bytes, including both header fields and data.

- For P.15, assume the bandwidth and the one-way propagation delay of the cross-country link is 100Mbps and 25ms, respectively.

$$\text{Utilization (User)} = (L / R) / (RTT + L / R)$$

$$L = 1500 \text{ bytes} * 8 \text{ bytes per packet} \\ = 12000$$

$$R(\text{trans rate}) = 100\text{Mbps}$$

"window" of up to $N \leq 2^k$ consecutive, unacknowledged packets

$$U > 98\%$$

$$D_{\text{prop}}(RTT) = 25\text{ms}$$

$$N(\text{window}) = ?$$

$$D(\text{trans}) = L/R \\ = 12000/(100*10^6)$$

$$= 0.00012$$

$$\text{Utilization (User)} = (L / R) / (RTT + L / R)$$

$$0.98 = N * (0.00012) / (.025 + 0.00012)$$

$$N = 205 \text{ packets}$$

- P22. Consider the GBN protocol with a sender window size of 4 and a sequence number range of 1,024. Suppose that at time t , the next in-order packet that the receiver is expecting has a sequence number of k . Assume that the medium does not reorder messages. Answer the following questions:

a. What are the possible sets of sequence numbers inside the sender's window at time t ? Justify your answer.

[0, 1, 2, 3], [k , $k + 1$], [K , $k+1$, $k+2$, $k+3$], [1021, 1022, 1023, 1024]

b. What are all possible values of the ACK field in all possible messages currently propagating back to the sender at time t ? Justify your answer.

[0, 1, 2, 3], [k , k], [k , $k + 1$, $k + 1$], [k , $k + 1$, $k + 2$, $k + 2$], [K , $k+1$, $k+2$, $k+3$], [1021, 1022, 1023, 1024]

- P25. We have said that an application may choose UDP for a transport protocol because UDP offers finer application control (than TCP) of what data is sent in a segment and when.

a. Why does an application have more control of what data is sent in a segment?

b. Why does an application have more control on when the segment is sent?

Flow control, when the sender keeps the amount of transmitted, unACKed data less than most recently received RcvWindow

In congestion control, the sender(s) adjust their transmission rate so as not to overwhelm routers in the network

- P27. Host A and B are communicating over a TCP connection, and Host B has already received from A all bytes up through byte 126. Suppose Host A then sends two segments to Host B back-to-back. The first and second segments contain 80 and 40 bytes of data, respectively. In the first segment, the sequence number is 127, the source port number is 302, and the destination port number is 80. Host B sends an acknowledgment whenever it receives a segment from Host A.

80 bytes

40 bytes

Seq 127

Source port 302

Dest port 80

- a. In the second segment sent from Host A to B, what are the sequence number, source port number, and destination port number?

Seq 207, source port 80, dest port 302

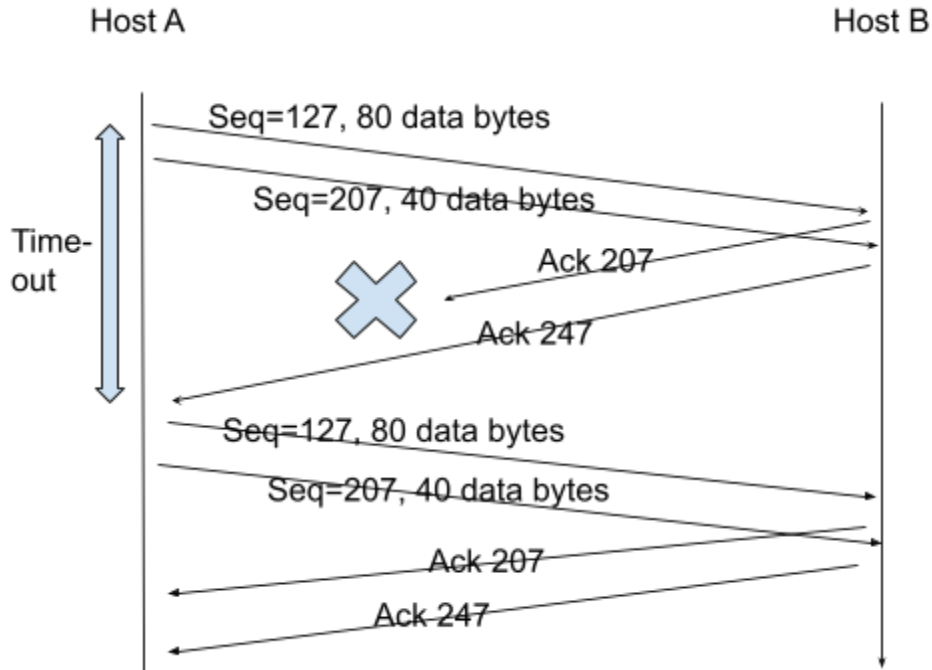
- b. If the first segment arrives before the second segment, in the acknowledgment of the first arriving segment, what is the acknowledgment number, the source port number, and the destination port number?

ACK 207, source port 80, dest port 302

- c. If the second segment arrives before the first segment, in the acknowledgment of the first arriving segment, what is the acknowledgment Number?

ACK 247

- d. Suppose the two segments sent by A arrive in order at B. The first acknowledgment is lost and the second acknowledgment arrives after the first timeout interval. Draw a timing diagram, showing these segments and all other segments and acknowledgments sent. (Assume there is no additional packet loss.) For each segment in your figure, provide the sequence number and the number of bytes of data; for each acknowledgment that you add, provide the acknowledgment number.



P37. Compare GBN, SR, and TCP (no delayed ACK). Assume that the timeout values for all three protocols are sufficiently long such that 5 consecutive data segments and their corresponding ACKs can be received (if not lost in the channel) by the receiving host (Host B) and the sending host (Host A) respectively. Suppose Host A sends 5 data segments to Host B, and the 2nd segment (sent from A) is lost. In the end, all 5 data segments have been correctly received by Host B.

- For P.37, only consider the comparison between GBN and TCP protocols only (do not consider SR protocol)

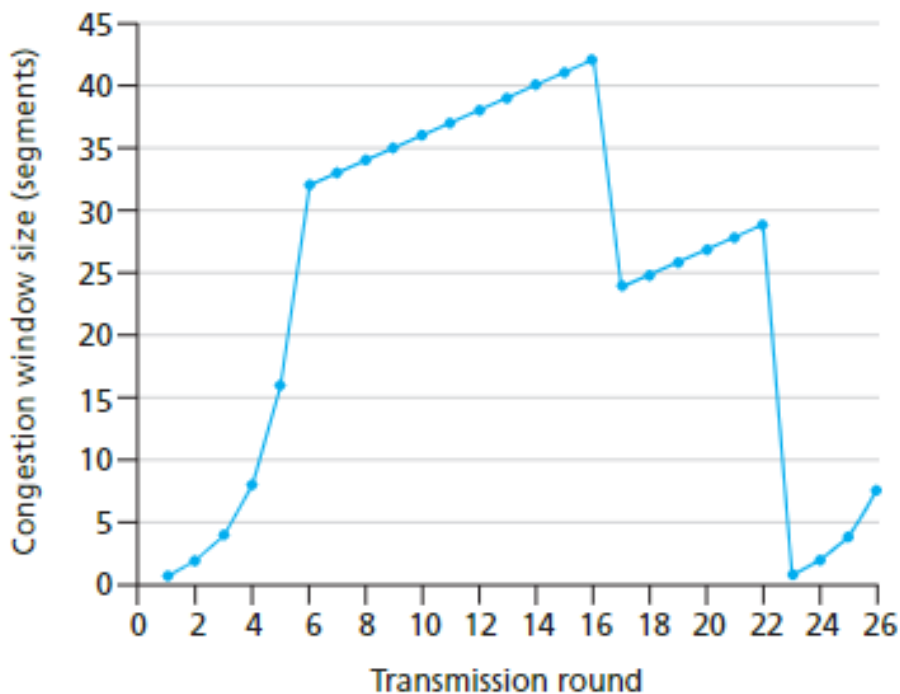
a. How many segments has Host A sent in total and how many ACKs has Host B sent in total? What are their sequence numbers? Answer this question for all three protocols.

GBN:

TCP:

b. If the timeout values for all three protocol are much longer than 5 RTT, then which protocol successfully delivers all five data segments in shortest time interval?

- P40. Consider Figure 3.58. Assuming TCP Reno is the protocol experiencing the behavior shown above, answer the following questions. In all cases, you should provide a short discussion justifying your answer.
- a. Identify the intervals of time when TCP slow start is operating.
 - b. Identify the intervals of time when TCP congestion avoidance is Operating.
 - c. After the 16th transmission round, is segment loss detected by a triple duplicate ACK or by a timeout?
 - d. After the 22nd transmission round, is segment loss detected by a triple duplicate ACK or by a timeout?
 - e. What is the initial value of ssthresh at the first transmission round?
 - f. What is the value of ssthresh at the 18th transmission round?
 - g. What is the value of ssthresh at the 24th transmission round?
 - h. During what transmission round is the 70th segment sent?
 - i. Assuming a packet loss is detected after the 26th round by the receipt of a triple duplicate ACK, what will be the values of the congestion window size and of ssthresh?
 - j. Suppose TCP Tahoe is used (instead of TCP Reno), and assume that triple duplicate ACKs are received at the 16th round. What are the ssthresh and the congestion window size at the 19th round?
 - k. Again suppose TCP Tahoe is used, and there is a timeout event at 22nd round. How many packets have been sent out from 17th round till 22nd round, inclusive?



- P43. Host A is sending an enormous file to Host B over a TCP connection. Over this connection there is never any packet loss and the timers never expire. Denote the transmission rate of the link connecting Host A to the Internet by R bps. Suppose that the process in Host A is capable of sending data into its TCP socket at a rate S bps, where $S = 10 \cdot R$. Further suppose that the TCP receive buffer is large enough to hold the entire file, and the send buffer can hold only one percent of the file. What would prevent the process in Host A from continuously passing data to its TCP socket at rate S bps? TCP flow control? TCP congestion control? Or something else? Elaborate
- P44. Consider sending a large file from a host to another over a TCP connection that has no loss.
- Suppose TCP uses AIMD for its congestion control without slow start. Assuming $cwnd$ increases by 1 MSS every time a batch of ACKs is received and assuming approximately constant round-trip times, how long does it take for $cwnd$ increase from 6 MSS to 12 MSS (assuming no loss events)?

b. What is the average throughput (in terms of MSS and RTT) for this connection up through time = 6 RTT?